

# Agentic BI

An AI-Powered Conversational Business Intelligence Platform

A multi-agent system for natural-language data analysis, automated visualization, and machine-learning-assisted business intelligence.

## Project Proposal

Prepared for academic project discussion

# 1. Project Overview

This project proposes an AI-powered Business Intelligence platform that allows users to interact with organizational data through natural-language conversations. Instead of requiring users to manually write queries, clean data, build dashboards, and select machine-learning techniques, the platform uses specialized AI agents to automate these activities.

A user could ask: "Why did our sales decrease last quarter, which regions were affected, and can you predict next month's sales?" The system would understand the request, retrieve and prepare the required data, perform the analysis, generate visualizations, and optionally build a machine-learning model to produce a forecast.

The platform will function as a conversational analytics assistant rather than simply displaying pre-built dashboards. Its central capability is to plan and execute an end-to-end analytical workflow based on the user's intent.

## 2. Problem Statement

Traditional BI tools often require users to manually connect data sources, understand data models, write queries, prepare datasets, create visualizations, and interpret the results. These steps can be difficult for non-technical users and can make analytics slow and dependent on specialized teams.

The proposed platform addresses this problem by allowing users to describe their analytical requirements in natural language. AI agents then coordinate data retrieval, preparation, analysis, visualization, and machine-learning tasks.

## 3. Main Objective

The objective is to develop a multi-agent BI platform that can:

- Understand business questions expressed in natural language.
- Retrieve and prepare relevant data for analysis.
- Perform descriptive, diagnostic, and predictive analysis.
- Automatically generate meaningful visualizations.
- Build and evaluate machine-learning models when required.
- Explain analytical results in an understandable and traceable manner.

## 4. Multi-Agent Architecture

The platform will be organized as a group of specialized agents. Each agent will be responsible for a specific stage of the analytics workflow, while the Orchestrator Agent coordinates the complete process.

### 0. Orchestrator Agent

Central coordinator of the platform.

- Understands the user's intent and maintains conversational context.
- Breaks complex questions into smaller analytical tasks.
- Decides which agents should be invoked and in what sequence.
- Combines agent outputs into a final response for the user.

### 1. Data Fetch/Integration Agent

Retrieves and prepares the data required for analysis.

- Identifies the required data sources and entities.
- Inspects schemas, metadata, files, tables, or API objects.
- Generates queries or API requests and retrieves the data.
- Handles authentication, pagination, retries, and connector errors.
- Cleans, normalizes, and merges related datasets.

### 2. Data Analytics Agent

Performs analytical operations on prepared data.

- Calculates KPIs, aggregations, and descriptive statistics.
- Performs trend, comparison, correlation, and segmentation analysis.
- Detects anomalies and outliers.
- Investigates possible causes behind observed business changes.

### 3. Visualization Agent

Converts analytical results into understandable visual outputs.

- Selects suitable chart types based on the analytical question.
- Generates charts, KPI cards, dashboards, and reports.
- Highlights important trends, comparisons, and anomalies.
- Supports follow-up questions and interactive exploration.

## 4. ML Agent

Handles machine-learning-related tasks when prediction or modeling is required.

- Identifies target and feature columns.
- Handles missing values, categorical features, scaling, and data leakage.
- Splits data and selects suitable algorithms.
- Trains and evaluates regression, classification, clustering, forecasting, or anomaly models.
- Generates predictions and explains important model results.

## 5. Example End-to-End Workflow

A user asks: "Which customers are at risk of leaving based on declining purchases and increasing support issues?"

| Step | Agent activity                                                                         |
|------|----------------------------------------------------------------------------------------|
| 1    | The Orchestrator Agent interprets the request and creates an execution plan.           |
| 2    | The Data Fetch/Integration Agent retrieves purchase and support data.                  |
| 3    | Related customer records are matched, cleaned, and merged.                             |
| 4    | The Data Analytics Agent identifies declining purchases and increasing support issues. |
| 5    | The ML Agent optionally builds a churn-risk model.                                     |
| 6    | The Visualization Agent generates customer-risk charts and summaries.                  |
| 7    | The Orchestrator Agent presents the final explanation in the chat interface.           |

The platform may respond: "Five customers show high churn risk. Their purchase frequency decreased by more than 20%, while unresolved support issues increased during the same period. Customer A has the highest risk because its revenue declined by 35% and its unresolved issues increased by 60%."

## 6. System Characteristics

- Conversational: users interact through natural-language questions and follow-ups.
- Multi-agent: specialized agents collaborate on different analytics tasks.
- Data-source independent: the workflow can work with different kinds of data sources.
- Predictive: the ML Agent can support forecasting and predictive modeling.
- Explainable: results include understandable reasoning and supporting evidence.
- Traceable: the system can record the data, transformations, queries, and methods used.
- Extensible: additional data sources, analytical methods, and agents can be added later.

## 7. Expected Outcome

The expected outcome is a functional conversational BI prototype that transforms natural-language questions into complete analytics workflows. The platform will demonstrate how AI agents can assist with data retrieval, data preparation, business analysis, visualization, machine learning, and insight generation.

## 8. Conclusion

The proposed project is an agentic Business Intelligence platform that combines conversational interaction with automated data analytics and machine learning. Instead of requiring users to manually navigate multiple BI tools, write queries, prepare datasets, select visualizations, and build models, the system allows them to describe their analytical requirement in natural language. The specialized agents then collaborate to execute the workflow and communicate the results in an understandable manner.

The project demonstrates a practical application of multi-agent AI in business analytics and aims to make advanced BI capabilities more accessible to non-technical users.